

MTH 310/520 Graph Theory: Final Exam

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Total: 30 points

This time the theme of the difficulty levels is coffee roast levels.



Fig. 1: Coffee roast levels

Instructions: Every student should attempt all five questions from Light Roast and the remaining points are to be attempted based on your course registration levels as mentioned below.

MTH310: Attempt the remaining 25 points from the remaining questions. In case you attempt more than the required number of points, you will be awarded points from the questions that fetched you maximum points. In case it so happens that the way you chose to answer questions gives you 5+5+5+10+10 attempted points, then you will not be graded out of 35 points. Rather, your points will be normalised to 30 points.

MTH520: Attempt the remaining 25 points by attempting any one question from Medium Roast, one from Dark Roast, and one from Chicory Blend. In case you attempt more than the required number of points, you will be awarded points from the questions that fetched you maximum points.

I. LIGHT ROAST

Light roasts produce fruity acidic coffee, great for beginners. A pourover is especially recommended. Just like light roasts, the problems in this level are easy, and are intended to freshen you up like a pourover, before you enter the more difficult levels.



Fig. 2: A pourover brewing setup

- I.i. Prove or disprove: Every graph with $|G|-1$ edges is a tree. [CO1]
- I.ii. Prove or disprove: If $\Delta(G) \leq 2$ then $L(G)$ is acyclic. [CO2]
- I.iii. Prove or disprove: A triangle-free graph is bipartite. [CO1]
- I.iv. Suppose a graph admits a ear decomposition. Is it necessarily planar? [CO3]
- I.v. Calculate $\chi'(Q_n)$ where Q_n is the n -hypercube graph. [CO3]

[1+1+1+1+1=5 points]

II. MEDIUM ROAST

Medium roasts are between light and dark roasts as the name suggests. They start to take on the bitterness one typically identifies with coffee, but continue to maintain their acidic flavour hinting at their fruity origins. Like medium roasts, this level of questions are not too dark, or rather difficult, but are not cakewalks like those from the previous levels.

II.i This set of questions will be about spanning trees. [CO1,CO3]

- a) Show that for any G with $\kappa(G) \geq 2k$, G can be packed by k spanning trees.
- b) Consider a graph G and call $F \subseteq E(G)$ to be *even* if every $v \in V(G)$ has an even number of edges from F incident on it. Show that if $F_1, F_2 \subseteq E(G)$ are even, then $F_1 \Delta F_2$ is even.
- c) Now, let $T \subseteq G$ be a spanning tree, and let $e \in E(G) \setminus E(T)$. Let $F(e)$ denote the edges of the fundamental cycle of e with respect to T . Show that $F(e)$ is even.
- d) Use parts b) and c) to show that there exists an even $F \subseteq E(G)$ such that $E(G) \setminus E(T) \subseteq F$.

[2+0.5+0.5+2=5 points]

II.ii This is a question on bipartite graphs. [CO3]

- a) Show that G is bipartite iff $\alpha(H) \geq \frac{|H|}{2}$ for every $H \subseteq G$.

[Hint: For the if part, observe that if G is not bipartite, then it must contain an odd cycle.]

- b) Show that a bipartite graph G possesses a matching of size at least $\frac{|G|\delta(G)}{2\Delta(G)}$.
- c) A bipartite graph is planar only if $|E(G)| \leq 2|G| - 4$.

[2+2+1=5 points]



Fig. 3: Yoshimura, the proprietor of Anteiku coffee shop in the anime *Tokyo Ghoul* making a pourover. You can see the gooseneck kettle which is used to pour water circularly over ground coffee.

II.iii This is a set on graph colouring. [CO3]

- a) Let G and H be graphs satisfying $V(G) \cap V(H) = \emptyset$. Then define the *join* of G and H as the graph

$$G + H := (V(G) \cup V(H), E(G) \cup E(H) \cup \{\{g, h\} : g \in V(G), h \in V(H)\}).$$

Show that $\chi(G + H) = \chi(G) + \chi(H)$.

- b) Let G be a triangle-free graph.

- i) Let $k := \lceil \frac{\Delta(G)+1}{4} \rceil$. Let $P = \{V_1, \dots, V_k\}$ be a partition of $V(G)$ which maximizes $|E(G|_P)|$. Let $G_i = G[V_i]$.

Show that $\Delta(G_i) \leq 3$.

[Hint: Assume WLOG a vertex $v \in V_1$ with degree at least 4 in V_1 . Can you show that there must exist some $i \neq 1$ such that $|N(v) \cap V_i| \leq 3$? Can you then construct a partition P' such that $|E(G|_{P'})| > |E(G|_P)|$?

- ii) Show that $\chi(G) \leq 3k$.

[Hint: Observe that $G \subseteq \sum_{i=1}^k G_i$.]

[1.5+2+1.5=5 points]

III. DARK ROAST

These are the stronger coffees exuding its bitterness while continuing to retain the flavours of the original beans. Perfect for espressos and ristrettos, and for those with a sweet tooth, affogatos. Like the dark roast, the questions at this level are strong and difficult.

III.i In this question we shall prove Mycielski's Theorem which states that for every $k > 0$ there exists a triangle free G with

$$\chi(G) = k. \text{ [CO3,CO4]}$$

We prove the theorem by induction. Define $M_k = K_k$ for $k \leq 2$ and observe that Mycielski's Theorem holds. Now assume that the theorem holds upto $k \geq 2$, let M_k be a graph satisfying the theorem. Let $V(M_k) = \{v_1, \dots, v_n\}$. Define

$$M_{k+1} := (V(M_k) \cup \{w_1, \dots, w_n, z\}, E(M_k) \cup \{\{w_i, v_j\} : \{v_i, v_j\} \in E(M_k)\} \cup \{\{z, w_i\} : i \in [n]\}).$$

- Show that M_{k+1} is triangle-free.
- Argue that $\chi(M_{k+1}) \geq \chi(M_k) = k$.
- Show that $\chi(M_{k+1}) \leq k + 1$.
- Now assume $\chi(M_{k+1}) = k$. Then, arrive at a contradiction by showing $\chi(M_k) < k$.

[Hint: Consider a k -colouring of M_{k+1} provided it exists. Observe that the colour on z cannot appear on any one of the w_i s. Can you use this fact to suitably recolour that vertices $v_1, \dots, v_n$?]

[3+1+2.5+3.5=10 points]



Fig. 4: A moka pot is the perfect machine for making espressos at home. I suggest using medium and darker roasts for this.

III.ii We shall prove here that any bipartite G satisfying $|A|=|B|=n$, $\delta(G) \geq 2$, and $\Delta(G) \leq 3$ has either no perfect matching or at least two perfect matchings. We shall prove by induction on the size of $|A|$. [CO3,CO4]

- Argue the base case where $n = 2$.
- Now assume the induction hypothesis holds up to $n - 1$. Next assume that G has a unique perfect matching. Show that there exists $S \subsetneq A$ such that $|N(S)| = |S|$.
[Be careful. You need to find $S \subsetneq A$. $S = A$ will always satisfy the requirement if a perfect matching exists.]
- Fix a set S of *minimal* size that satisfies part b). Let k be the number of degree two vertices in S and let l be the number of degree two vertices in $N(S)$. Show that $k \geq l$ and there are exactly $k - l$ edges from $N(S)$ to $A \setminus S$.
- Using the minimality of S , show that every $b \in N(S) \cap N(A \setminus S)$ is of degree 3. Furthermore, exactly one of the three edges incident with b is connected to a vertex in $A \setminus S$.
- Argue that $G[S \cup N(S)]$ has a perfect matching, and hence conclude the proof of the induction step by arriving at a contradiction.

[0.5+2+2.5+2.5+2.5=10 points]

IV. CHICORY BLEND

South India is the oldest coffee culture of the subcontinent where coffee is believed to be introduced by the sufi saint Baba Budan. Baba Budan giri, named after the saint, as well as the surrounding hills of the Malnad/Malenadu, are today dotted with coffee plantations. In order to reduce the price of what had become an everyday drink, South Indian blenders and roasters began to mix coffee with chicory root, a low cost substitute. The blend, while initially meant to reduce costs, produced a whole new flavour, and gave South Indian coffee its distinct identity. Just like the chicory blend, this level mixes graph theory with other fundamentals of mathematics to create a whole new range of flavours and theorems.

IV.i. Let $G = (V, E)$ be a bipartite graph with $V = A \cup B$. [CO4]

- Uniformly and independently place each vertex $v \in B$ into S with probability $1/2$. Then, for every vertex $v \in A$, include v in S if and only if $|N(v) \cap S|$ is odd. Argue that this construction ensures that $d_{G[S]}(v)$ is odd for every $v \in A \cap S$.
- Let $X_v := \mathbf{1}\{v \in S\}$. Calculate $P[X_v = 1]$ for $v \in A$.
- Argue via the expectation of $|S|$ that there exists an S with $|S| \geq \frac{n}{2}$.
- Give a formal statement of the result we proved using parts a)-c).

[2+4+2+2=10 points]



Fig. 5: South Indian coffee is generally had with milk and sugar in brass or stainless steel glass and tumbler setups. The brewer is a multidrip brewer, similar to a Vietnamese Phin brewer.

IV.ii. Recall from Assignment 1 that given any $n \times n$ real symmetric matrix M , we showed that $\lambda_{\max}(M) = \max_{x \neq 0} \frac{x^T M x}{||x||^2}$, where $\lambda_{\max}(M)$ denotes the largest eigenvalue of M . For $i \in [n]$ define M_i to be the i th principal submatrix of M , i.e., the matrix obtained by deleting the i th row and column of M .

Let G be a graph, and let $\lambda_1 := \lambda_{\max}(A(G))$. Recall from Assignment 1 that $\lambda_1 \geq d(G)$. In this question we shall prove Wilf's Theorem which states that $\chi(G) \leq \lfloor \lambda_1 \rfloor + 1$. [CO3,CO4]

a) Show that M_i is an $(n-1) \times (n-1)$ real symmetric matrix.

b) For every $i \in [n]$, show that $\lambda_{\max}(M) \geq \lambda_{\max}(M_i)$.¹

[Hint: For every $x \in \mathbb{R}^n$, consider the vector $x_{i,0}$ defined as $x_{i,0}(j) = x(j)$ for all $j \neq i$ and $x_{i,0}(i) = 0$. What can you say about $x_{i,0}^T M x_{i,0}$?]

c) We shall now prove Wilf's Theorem using induction on $|G|$. First prove the Theorem for the base case of $|G|=1$.

d) Now, assume that the induction hypothesis is valid for all graphs of order at most $|G|-1$. Argue that there exists a $v \in V(G)$ satisfying $d(v) \leq \lfloor \lambda_1 \rfloor$.

e) Conclude the proof of the induction step.

[Hint: Let $S = V(G) \setminus \{v\}$. What can you say about $\chi(G[S])$ using the induction hypothesis?]

[1+3+0.5+1.5+4=10 points]

¹This is a tiny chunk of a big result called *Cauchy's interlacing formula* to prove which one needs a general version of $\lambda_{\max}(M) = \max_{x \neq 0} \frac{x^T M x}{\|x\|^2}$ referred to as the *Courant-Fischer Theorem*.